Genuine Multipartite Entanglement Analysis

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Model
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Hubbard ring (1-D, PBC)

System parameters
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Sites (L)         : 4
Electrons (Ne)    : 4
Hopping (t)       : 1.0
Interaction (U)   : 1.0
Mode              : uniform
n-body order      : 4
```

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Ground-state properties
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E0                : -3.34084762
E1                : -3.29295138

<S2>              : 0.000000
S                 : 0.0000
<Sz>              : 0.000000
Multiplicity 2S+1 : 1.0
Spin state        : Singlet (S=0)

Double occupancy
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<D> total         : 0.577987
<D> / site        : 0.144497
```

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GME summary
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Extracted n-spin RDM : n = 4

Purity:
  gamma = Tr(rho^2)

Separable cut criterion:
  gamma_A = 1  <->  rho = rho_A x rho_B

C_GME (Troiani et al.):
  C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Decision logic:
  gamma=1, all Tr(rho_A^2)<1  -> GME
  gamma=1, some Tr(rho_A^2)=1 -> No GME
  gamma < 1                   -> Mixed

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